

Aurora Home — Installation and Setup Guide

Thermostat T3 · Sensor S1 · Hub H2 — firmware 4.2.1 · document revision 7

1. About this guide

This guide covers the physical installation and first-time setup of the Aurora Home range: the T3 thermostat, the S1 door and window sensor, and the H2 hub. It is written for homeowners doing their own installation and for installers commissioning several properties at once. Where a step requires an electrician, the text says so explicitly.

Read section 2 before opening the packaging. The T3 connects to mains-derived low-voltage wiring, and the sequence in which you isolate the boiler matters.

2. Safety and prerequisites

Isolate the heating system at the consumer unit before removing an existing thermostat. Confirm the circuit is dead with a voltage tester rather than relying on the boiler's own display, which can remain lit from a separate supply. Aurora Home devices are for indoor use only, in dry rooms, at ambient temperatures between 0 °C and 45 °C.

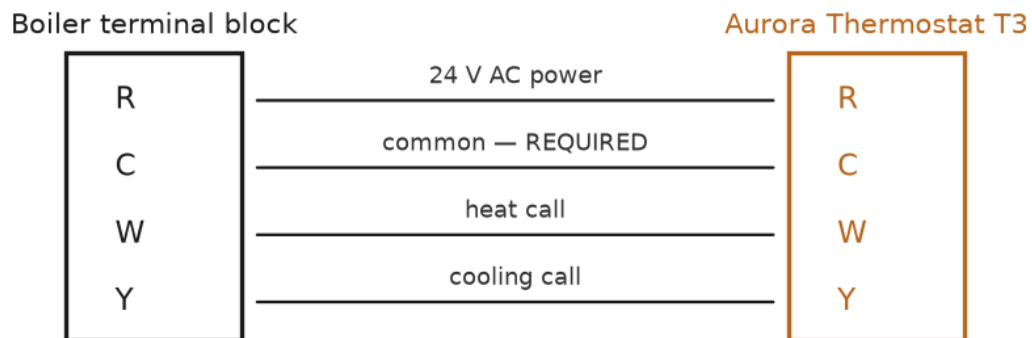
You will need: a small flat screwdriver, a voltage tester, a drill with a 6 mm masonry bit if you are mounting on brick, and a smartphone running the Aurora Home app. Allow forty minutes for a single-thermostat installation and around two hours for a whole-property commissioning.

Note: if your existing thermostat is battery-powered and has only two wires, your system almost certainly lacks the common conductor the T3 needs. Read section 3 before ordering.

3. Wiring the T3

The T3 replaces an existing wall thermostat and reuses its backplate position in most installations. Remove the old faceplate, photograph the terminal block before disconnecting anything, and label each conductor as you release it. The terminal designations used by Aurora Home follow the common four-wire convention.

The diagram below shows the mapping between a typical boiler terminal block and the T3 backplate. Match the labels rather than the colours: conductor colours vary by installer and by country, and are not a reliable guide.



The T3 draws continuous power: the C terminal must be connected.

Figure 1: Terminal mapping between a four-wire boiler block and the T3 backplate.

Tighten each terminal to a firm stop, then tug each conductor gently to confirm it is captive. Refit the backplate, clip the T3 body on until it clicks, and restore power at the consumer unit. The display illuminates within ten seconds.

3.1 If the display stays dark

A dark display after restoring power almost always means the backplate is not seated or a terminal has not made contact. Isolate the circuit again, remove the body, and inspect the four spring contacts on the rear for debris or bent leaves. Only after that should you suspect the unit itself.

4. Commissioning the H2 hub

The H2 hub bridges the Aurora radio mesh to your home network. It must sit within radio range of at least one powered device, and within cable reach of your router. The hub has no battery: it is powered permanently from the supplied adapter, and a power cut interrupts automation until the hub rejoins the network.

Connect the hub to the router with the supplied cable, wait for the status ring to settle to a steady white, then open the Aurora Home app and choose Add hub. The app discovers hubs on the same subnet automatically. If discovery fails, the hub can be added manually using the twelve-character identifier printed on its base.

4.1 Pairing devices

Devices pair one at a time. Put the hub into pairing mode from the app, then hold the pairing button on the device for three seconds until its indicator pulses. Pairing completes within thirty seconds; the device then appears in the app under its default name, which you should change immediately to something that identifies the room. Automations refer to devices by name, and renaming a device later does not rewrite existing automations.

Sensors join the mesh as leaf nodes and do not relay traffic for other devices. Thermostats and any mains-powered accessory act as repeaters, so in a large property the practical range depends far more on where the thermostats are than on where the hub is.

4.2 Placement and range

Radio range in free air is generous, but real buildings are not free air. Foil-backed insulation, wet plaster and reinforced concrete all attenuate the signal sharply. As a rule of thumb, allow one internal wall between any sensor and its nearest repeater, and two at the absolute limit. Where a sensor sits at the far end of a long extension, adding a mains-powered accessory in between is more reliable than moving the hub.

Avoid mounting the hub inside a metal cabinet or directly beside the router: the router's own radios

raise the noise floor and reduce effective mesh range.

5. Zoning and schedules

A zone groups one thermostat with any number of sensors. The thermostat holds the schedule; the sensors report temperature and occupancy into it. Where a zone contains several sensors, the thermostat averages their readings by default, and can be configured instead to follow whichever sensor reports occupancy.

Schedules are per zone and per day, with up to six setpoints in each day. A setpoint applies until the next one begins; there is no explicit off period. To leave a room unheated overnight, set a low setpoint rather than deleting the evening entry, which would extend the previous setpoint until morning.

Holiday mode overrides every schedule in the property with a single frost-protection setpoint, and ends automatically on the return date you set.

6. Compatibility matrix

The matrix below lists the heating system types verified by Aurora Home for the current firmware. Systems marked as requiring an interface module need the AH-IM1 accessory between the T3 and the boiler; systems marked as unsupported cannot be controlled by the T3 at all and are not made supportable by an accessory.

Verified means Aurora Home has tested the combination in its own laboratory. Combinations not listed may still work, but are not covered by support.

System type	Control method	T3	Interface module	Notes
Combi boiler, on/off	Volt-free contact	Verified	Not required	Most common UK and DE installations
Combi boiler, OpenTherm	Modulating	Verified	Not required	Modulation requires firmware 4.1 or later
System boiler with cylinder	Volt-free contact	Verified	Not required	Hot water control via separate channel
Heat-only boiler	Volt-free contact	Verified	Not required	Pump overrun handled by the boiler
Underfloor heating, wet	Zone valve	Verified	Required	One valve actuator per zone
Underfloor heating, electric	Direct switching	Not supported	—	Load exceeds the T3 relay rating
District heating substation	0–10 V	Verified	Required	Set the actuator range in the app
Air-source heat pump, on/off	Volt-free contact	Verified	Not required	Minimum cycle time must be set to 15 minutes
Air-source heat pump, modulating	Proprietary bus	Not supported	—	No open control interface
Ground-source heat pump	Volt-free contact	Verified	Required	Interface module isolates the control circuit
Electric panel heaters	Direct switching	Not supported	—	Load exceeds the T3 relay rating
Storage heaters	Time-of-use relay	Not supported	—	Charge control is outside the T3's remit
Biomass boiler	Volt-free contact	Verified	Required	Interface module required for safety interlock
Oil boiler, on/off	Volt-free contact	Verified	Not required	Same wiring as a heat-only boiler
Warm-air unit	Volt-free contact	Verified	Not required	Fan delay configured on the unit itself
Solar thermal, supplementary	Monitoring only	Verified	Not required	T3 reads, never controls, the solar circuit
Radiator valves, third-party	Radio	Not supported	—	Only Aurora Home valves join the mesh
Ventilation with heat recovery	Volt-free contact	Verified	Required	Boost function mapped to a schedule
Gas absorption heat pump	Volt-free contact	Verified	Required	Interface module handles the interlock
Hybrid boiler and heat pump	OpenTherm	Verified	Not required	The hybrid controller decides which source runs

System type	Control method	T3	Interface module	Notes
Wood pellet stove with back boiler	Volt-free contact	Verified	Required	Manual override on the stove takes precedence
Electric boiler, wet system	Volt-free contact	Verified	Not required	Check the relay rating against the boiler contactor
Two-pipe fan coil units	0–10 V	Verified	Required	One actuator per coil
Four-pipe fan coil units	0–10 V	Not supported	—	Simultaneous heating and cooling is not modelled
Radiant ceiling panels	Zone valve	Verified	Required	Slow response; widen the deadband
Towel rail, electric	Direct switching	Not supported	—	Load exceeds the T3 relay rating
Immersion heater	Time-of-use relay	Not supported	—	Hot water only; outside the T3's remit
Thermal store with coil	Volt-free contact	Verified	Not required	Store temperature read by a separate sensor
Air handling unit, commercial	0–10 V	Not supported	—	Commercial plant is out of scope
Infrared panel heaters	Direct switching	Not supported	—	Load exceeds the T3 relay rating
Condensing boiler, weather compensated	OpenTherm	Verified	Not required	Compensation curve stays on the boiler
Multi-zone manifold, six zones	Zone valve	Verified	Required	One interface module per manifold
Swimming pool heat exchanger	Volt-free contact	Not supported	—	Outside the tested temperature range

Table 1: Verified heating system types for firmware 4.2.1. The table continues across the page break; the header row repeats.

7. Capacity limits

A single H2 hub supports a maximum of 64 paired devices, counting thermostats, sensors and accessories together. Properties needing more capacity add a second hub; the app presents multiple hubs as one property, and automations may span them. Each hub maintains its own mesh, and a device belongs to exactly one hub.

In practice the radio mesh becomes the limiting factor before the device count does. Above roughly forty devices, Aurora Home recommends at least three mains-powered repeaters spread across the property.

8. Firmware updates

The hub checks for firmware nightly and stages updates for connected devices. Thermostats update in place and are unavailable for roughly ninety seconds. Sensors update opportunistically when they next report, so a full property can take several days to converge — this is expected and needs no intervention.

Updates cannot be rolled back. If an update introduces a problem in your specific installation, contact support rather than attempting to reflash a device.

9. Decommissioning

Remove devices from the app before removing them from the wall. A device removed physically but left paired continues to occupy a slot against the 64-device limit and will be reported as unreachable. If you have already removed the hardware, the app offers a forced removal that clears the pairing from the hub's side.

Factory-resetting the hub erases every pairing, all schedules and all automation history. There is no export, so record any complex schedule before resetting.

10. Commissioning several properties

Installers commissioning a block of flats or a small estate should treat each dwelling as a separate property in the app rather than as zones of one large property. Zones within a property share holiday mode and share the device budget of their hub, neither of which is what you want across separate households. The extra work is a few minutes per dwelling at handover and saves an awkward migration later.

Name devices for the room rather than the resident. Residents change; rooms do not, and a schedule that refers to a former tenant's name is confusing for whoever inherits it. Where a property has several identical rooms, add a floor prefix so that the sorted device list matches the order you would walk the building in.

10.1 Handover to the resident

Hand over the hub identifier in writing and confirm the resident can sign in before you leave. An installer account that retains access after handover is a support liability: the resident cannot remove it themselves, and Aurora Home will not remove it on request without the installer's confirmation. Transfer ownership properly at handover rather than sharing a login.

Leave the property with at least one schedule in place. A system with no schedule holds whatever setpoint was last entered, which after commissioning is usually a test value rather than something anyone wants to live with.

11. Maintenance

Aurora Home devices need no scheduled servicing. Sensor batteries are the only consumable, and the app reports a low battery well before the device stops reporting. Replace with the specified cell type: rechargeable equivalents have a lower nominal voltage and the low-battery warning will fire almost immediately.

Clean the thermostat with a dry cloth. Do not use solvents on the display bezel, and do not spray any cleaner directly at the unit — the vent slots on the underside feed the temperature sensor, and liquid drawn in there will skew readings long before it does any visible damage.

11.1 Seasonal checks

Before the heating season, run each zone for ten minutes and confirm the expected radiators warm up. Zone valves that have sat unused all summer occasionally stick, and finding that in September is considerably better than finding it in December. The app's zone test does this in sequence and reports which zones responded.

After the heating season, leave the system powered. Holiday mode and frost protection both depend on the hub being online, and a system switched off at the consumer unit protects nothing.

12. Glossary

Zone — one thermostat and the sensors associated with it, sharing a single schedule. **Setpoint** — the target temperature from a given time until the next setpoint begins. **Repeater** — any mains-powered device that relays mesh traffic on behalf of others. **Leaf node** — a battery-powered device that does not relay. **Deadband** — the difference between the temperature at which heating starts and the temperature at which it stops, which prevents rapid cycling. **Interface module** — the AH-IM1 accessory, required where the boiler cannot be switched directly by the T3 relay.